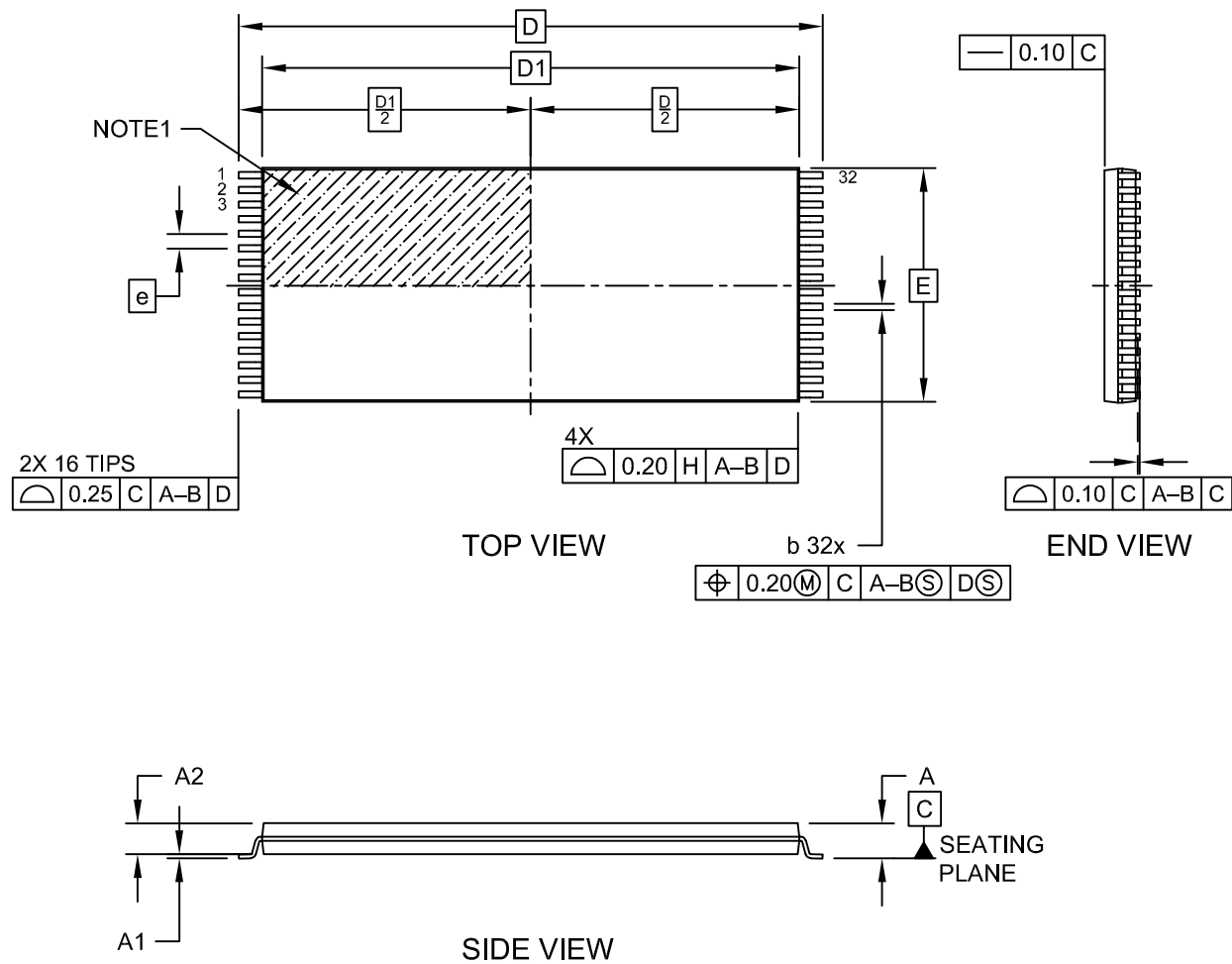


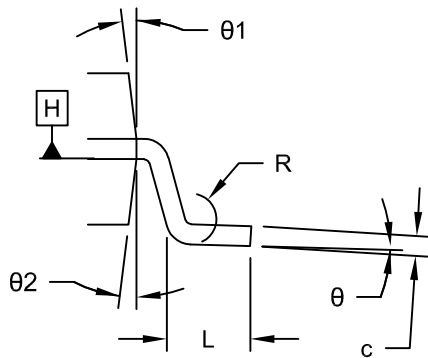
32-Lead Plastic Thin Small Outline Package (NLB) - 8x18.4x1.0 mm Body [TSOP] Atmel Legacy Package (TCS)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

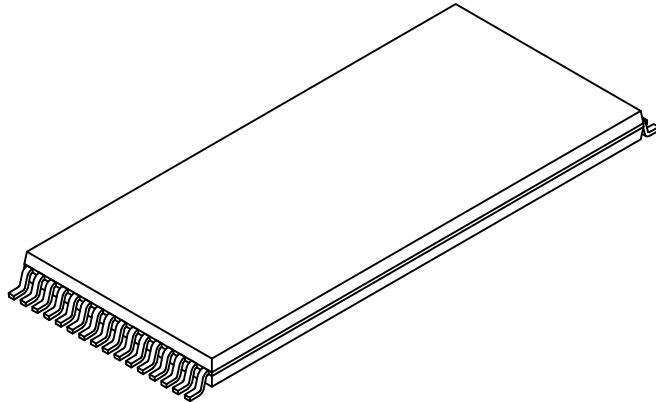


32-Lead Plastic Thin Small Outline Package (NLB) - 8x18.4x1.0 mm Body [TSOP] Atmel Legacy Package (TCS)

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SECTION A-A



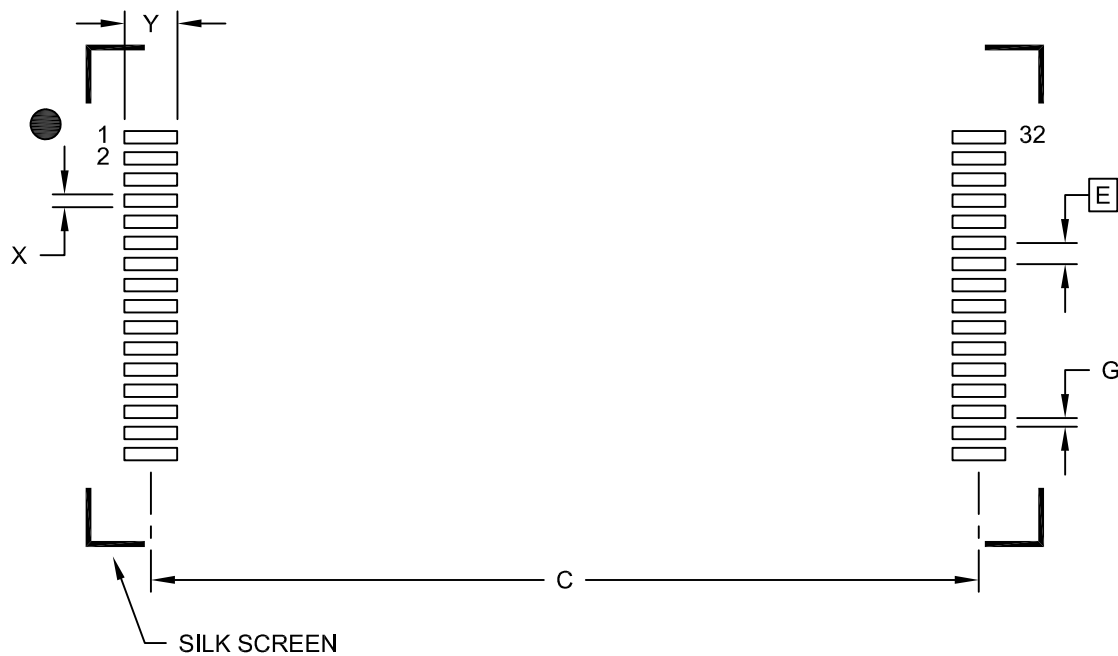
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	32		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.05	-	0.15
Molded Package Thickness	A2	0.95	1.00	1.05
Overall Length	D	20.00 BSC		
Molded Package Length	D1	18.40 BSC		
Overall Width	E	7.90	8.00	8.10
Terminal Width	b	0.17	0.22	0.27
Terminal Thickness	c	0.10	-	0.21
Terminal Length	L	0.50	0.60	0.70
Lead Bend Radius	R	0.127	-	-
Foot Angle	θ	0°	-	5°
Mold Draft Angle	θ1	6°	7°	8°
Mold Draft Angle	θ2	6°	7°	8°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

32-Lead Plastic Thin Small Outline Package (NLB) - 8x18.4x1.0 mm Body [TSOP] Atmel Legacy Package (TCS)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C		19.60	
Contact Pad Width (X32)	X			0.30
Contact Pad Length (X32)	Y			1.25
Contact Pad to Center Pad (X30)	G	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23343 Rev A